SYDNEY SMITH, PhD

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RESEARCH PROFILE

I study how neural systems transform dynamic sensory information into adaptive behavior, with a particular focus on how this transformation is implemented by physiologically plausible patterns of neural activity within and across brain systems. To this aim, I develop and apply approaches that preserve a link between measurable neural signals and the physiological mechanisms that generate behavior.

RESEARCH POSITIONS

Jul 2025–Present **Postdoctoral Associate** — Brain & Cognition Lab, Yale University; Anna Christina (Kia) Nobre

- Research focus: non-invasive and intracranial human neurophysiology to characterize brain networks and dynamics in adaptive cognition.

Sep 2019–Mar 2025 **PhD Candidate** — Neural and Data Analytics Lab, University of California, San Diego; Bradley Voytek

- Research: oscillatory basis of dynamic auditory attention using intracranial EEG; aperiodic mechanisms of seizure-based therapies using EEG in collaboration with San Diego VA Health.

Apr 2018–Aug 2019 **Lab Manager** — Neural and Data Analytics Lab, University of California, San Diego; Bradley Voytek

- Research: mechanisms of electroconvulsive therapy for major depressive disorder using EEG; phase preference in visual perceptual learning using EEG.

Aug 2016–Jan 2018 **Undergraduate Research Assistant** — Action Lab, University of California, Santa Barbara; Scott Grafton

- Research: approach–avoidance decision-making using fMRI; intentional binding using TMS.

EDUCATION

Mar 2025 **PhD in Neurosciences** — Specialization in Computational Neuroscience, University of California, San Diego

- Advisor: Bradley Voytek
- Dissertation committee: Eric Halgren, Gal Mishne, Greg Appelbaum
- Dissertation: “Gnarly Waves: Honoring neurophysiology in human cognitive and clinical neuroscience”
- **Chancellor’s Dissertation Medal**

Jan 2018 **BS in Biopsychology** — Minor in History of Art and Architecture, University of California, Santa Barbara

- Dean’s Honors

GRANTS & FUNDING

2025–Present **Swebilius Foundation Grant** — Yale University School of Medicine Comprehensive Epilepsy Center

- Project: Evaluating the utility of aperiodic connectivity in characterizing focal epilepsy
- Amount: $25,000
- Role: PI

HONORS & AWARDS

- Kavli Foundation Science & Society Sparks (2026; $7,500)
- UC San Diego Chancellor’s Dissertation Medal, School of Medicine (2026)
- NIH Outstanding Scholars in Neuroscience Award Program, National Institutes of Mental Health (OSNAP) (2024)
- UC San Diego Leon Thal Award for Excellence in Neuroscience Research (2024; $500)
- UC San Diego Lab Expo Grad Showdown public speaking competition, Winner (2023)
- UC San Diego Malcolm R. Stacey Memorial Fellowship (2021; $5,000)

PUBLICATIONS

Peer-Reviewed Publications

- Cellier D, **Smith SE**, Preston M, van Engen Q, Kosik-Rose E, Bender A, Hammonds R, Voytek B. (2026). Time-resolved spectral parameterization: recommendations and guidelines. *Neuron*. https://doi.org/10.1016/j.neuron.2026.08.009
- Preston M, **Smith SE**, Voytek B. (2026). Potential mechanisms and functional significance of aperiodic neural activity. *Nature Human Behavior*. https://doi.org/10.1038/s41562-026-02503-7
- Stolz L, Kohn J, **Smith SE**, Benster L, Appelbaum L. (2023). Predictive biomarkers of treatment response in Major Depressive Disorder. *Brain Sciences*. https://doi.org/10.3390/brainsci13111570
- **Smith SE**, van Engen Q, Kosik EL, Kohn J, Hill AT, Zomorrodi R, Blumberger DM, Daskalakis ZJ, Hadas I, Voytek B. (2023). Magnetic seizure therapy and electroconvulsive therapy increase aperiodic activity. *Translational Psychiatry*. https://doi.org/10.1038/s41398-023-02631-y
- **Smith SE**, Ma V, Gonzalez C, Chapman A, Printz D, Voytek B, Soltani M. (2023). Clinical EEG slowing induced by electroconvulsive therapy is better described by increased frontal aperiodic activity. *Translational Psychiatry*. https://doi.org/10.1038/s41398-023-02634-9 (**media coverage** — see Public Scholarship)
- Mercier MR et al., including **Smith SE** and Voytek B. (2022). Advances in human intracranial electroencephalography research, guidelines, and good practices. *NeuroImage*. https://doi.org/10.1016/j.neuroimage.2022.119438
- Waschke L, Donoghue T, Fiedler L, **Smith S**, Garrett DD, Voytek B, Obleser J. (2021). Modality-specific tracking of attention and sensory statistics in the human electrophysiological spectral exponent. *eLife*. https://doi.org/10.7554/eLife.70068

Commentaries

- **Smith SE**. (2025). Analyzing the elephant: The search for electroencephalography biomarkers. *Biological Psychiatry: Global Open Science*. https://doi.org/10.1016/j.bpsgos.2025.100459

Manuscripts Under Review

- **Smith SE**, Cellier D, Rigby A, Rosen BQ, Garrett J, Simon AJ, Shih J, Ben-Haim S, Doelling KB, Voytek B. Simulated field potentials capture multiple simultaneous processes underlying human temporal attention (preprint). *bioRxiv*. https://doi.org/10.64898/2026.09.17.752423. In review, *Nature Neuroscience*.

INVITED TALKS & WORKSHOPS

- **Smith SE** et al. (2023). Spectral parameterization of EEG reveals a plausible biological mechanism for therapeutic effects of seizure-based treatments for depression. Invited presentation, IEEE Conference on Signals, Systems, & Computers.
- **Smith SE** et al. (2023). Methods for Analyzing Neural Oscillations and Aperiodic Activity. **Workshop Leader**, Society for Psychophysiological Research.
- Invited lectures and seminars: CCBBI Neuroimaging Trainee Group, The Ohio State University (2026) — Brain Stimulation Group, Yale University (2025) — Brain and Cognition Lab, Yale University (2024) — Neurosciences Program Retreat, UC San Diego (2024) — Cognition at the Shore, UC San Diego (2024) — Neurosciences Interview Week Seminar, UC San Diego (2024) — PRESIDIO Group, UCSF (2024) — Knight Lab, UC Berkeley (2024) — Appelbaum Lab, UC San Diego (2022) — NeuroTechX, UC San Diego (2020).

CONFERENCE PRESENTATIONS

- **Smith SE**, Calbick D, Nobre AC, McCarthy G. (2026). Evaluating aperiodic-driven brain network connectivity in visual target detection. Poster, Organization for Human Brain Mapping.
- Vairagare I, Chapman A, **Smith SE**, Voytek B. (2025). Aperiodic activity as an EEG biomarker for sertraline response: Applications and limitations. Poster, Society for Neuroscience.
- **Smith SE**, Doelling K, Voytek B. (2024). Modeling the independence and convergence of frontal top-down anticipation signals and bottom-up auditory drive in the human auditory cortex. Poster, Society for Neuroscience.
- **Smith SE**, Doelling K, Voytek B. (2023). Investigating the mechanisms of precise predictive timing: simulation and intracranial EEG. Poster, Society for Neuroscience.
- **Smith SE**, Rigby A, Doelling K, Voytek B. (2023). Investigating the mechanisms of precise predictive timing: behavior and intracranial EEG. Poster, Cognitive Neuroscience Society.
- Chapman A, **Smith SE**, Voytek B. (2022). Hemispheric differences in aperiodic activity interact with symptom chronicity in Major Depressive Disorder to predict Sertraline treatment response. Poster, Society for Neuroscience.

- VanEngen Q, Kosik E, **Smith SE**, Voytek B. (2022). Periodic and aperiodic mechanisms of electroconvulsive therapy for major depressive disorder. Poster, Cognitive Neuroscience Society.
- **Smith SE**, Printz D, Voytek B, Soltani M. (2021). Increases in aperiodic activity explain EEG "slowing" after electroconvulsive therapy. Poster, Society for Neuroscience.
- **Smith S**, Ma V, Jacobson M, Printz D, Voytek B, Soltani M. (2019). Oscillatory mechanisms of electroconvulsive therapy for major depressive disorder. Poster, Society for Neuroscience.
- Waschke L, Donoghue T, **Smith S**, Voytek B, Obleser J. (2019). Aperiodic EEG activity tracks 1/f stimulus characteristics and the allocation of cognitive resources. Poster, Cognitive Computational Neuroscience.

TEACHING

- Certificate of College Teaching Preparation for Postdocs, Yale University (2026-present).
- Neuroanatomy, graduate-level laboratory teaching assistant, UC San Diego (2022–2023).
- Workshop Lead: Methods for Analyzing Neural Oscillations and Aperiodic Activity, Society for Psychophysiological Research (2023).

MENTORING

Undergraduate Research Mentorship

- Bridget Nussbaum — Biomedical Engineering, Yale University (2025–Present)
- Genevieve Kim — Neuroscience, Yale University (2026)
- Indrani Vairagare — Cognitive Science / Machine Learning, UC San Diego (2024–2025)
- Sarah Rosemay — Cognitive Science / Neuroscience, UC San Diego (2023–2024); current: Medical Coordinator/Consultant
- Angela Chapman — Clinical Psychology, UC San Diego (2020–2023); **Awards**: Barry M. Goldwater Scholarship, McNair Scholar, and NSF Graduate Research Fellowship; current: Clinical Psychology PhD, University of Iowa
- Shuangquan Feng — Cognitive Science / Machine Learning, UC San Diego (2020–2021); current: Neurosciences PhD, UC San Diego
- Chase Oden — Data Science, UC San Diego (2020–2021); current: Consultant, Deloitte
- Valentina Carreno — Cognitive Science / Machine Learning, UC San Diego (2020–2021); current: Health Tech Consultant, Oracle
- Lulu Ricketts — Data Science, UC San Diego (2020–2021); current: Perception Engineer, Aurora
- Celene Gonzales — Summer Training Academy for Research Success, UC San Diego (2018–2019); current: Clinical Research Coordinator, UC San Diego
- Meyhaa Buvanesh — Cognitive Science / Machine Learning, UC San Diego (2018–2019); current: Product Manager, Hippocratic AI
- Lakshmi Menon — Cognitive Science / Machine Learning, UC San Diego (2018–2019); **Awards**: DeepMind Fellowship and NYU Center for Data Science recognition; current: Software Test Engineer, Qualtrics
- Vincent Ma — Post-baccalaureate research, San Diego VA Health System (2018–2019); current: MD resident, Los Angeles General Medical Center

ACADEMIC SERVICE & PROFESSIONAL ACTIVITIES

Ad hoc peer review

- Progress in Neurobiology — PLOS Computational Biology — Imaging Neuroscience — Biological Psychiatry: Global Open Science — Addiction Biology.

Committee Service

- Fellow, Wu Tsai Institute SPPC, Yale University (2025–2026).
- Interview Week Committee, Neurosciences Graduate Program, UC San Diego (2020–2024).
- Admissions Committee, Neurosciences Graduate Program, UC San Diego (2020–2022).
- Career Development Committee, Neurosciences Graduate Program, UC San Diego (2019–2023).

PUBLIC SCHOLARSHIP & SCIENCE COMMUNICATION

Authored Public Scholarship

- Smith SE. (2023). *How electroconvulsive therapy heals the brain – new insights into ECT, a stigmatized yet highly effective treatment for depression*. The Conversation. [link]

Science Communication and Outreach

- Regional Director / Founder, Two Scientists Walk Into a Bar, New Haven (2026–Present).
- Writer / Editor, Stories of Women in Neuroscience (2021–2026); selected profiles include Dr. Farah Lubin [link] and Dr. Liberty Hamilton [link].
- City Coordinator, Pint of Science Festival, New Haven (2026).
- Science Communicator, Two Scientists Walk Into a Bar, San Diego (2024–2025).
- Invited Speaker, Nerd Nite San Diego (2024).
- Consultant, National Academy of Sciences Science & Entertainment Exchange (2024).
- Invited Speaker, SWENext, Pacific Ridge High School (2024).
- Invited Speaker, Pacific Ridge High School Biotechnology class (2023, 2025).
- Event Coordinator, Taste of Science Festival, San Diego (2020, 2022–2024).
- Volunteer, Brain Bee, San Diego (2020, 2022, 2024).
- Panelist, How to Find an RA Position & Succeed in the Lab, UC San Diego Career Center (2023).
- Volunteer Judge, International Youth Neuroscience Association NEURON competition (2020).

Selected Media Coverage

- Quanta Magazine: Brain's "background noise" may explain value of shock therapy. [link]
- Psychiatric News: Brain's "background noise" changes with ECT, MST.
- NPR/KPBS: Electroshock therapy works and now we may know why. [link]
- ScienceAlert: Brain's Background Noise Could Explain Electroconvulsive Therapy's Success.
- US News & World Report: Scientists May Have Solved Mystery of How Shock Therapy Works.
- UC San Diego Today: New Studies of Brain Activity Explain Benefits of Electroconvulsive Therapy.

ADDITIONAL TRAINING & PROFESSIONAL DEVELOPMENT

- OSL Workshop, BrainWorks at Wu Tsai Institute, Yale University (2025): Advanced MEG processing and analysis methods from the Oxford Center for Human Brain Activity Software Library.
- OPM-MEG Flux Workshop, BrainWorks at Wu Tsai Institute, Yale University (2025): Theory and practice of optically pumped magnetometer MEG in cognitive neuroscience research.
- MicroMBA, Rady School of Management, UC San Diego (2022).
- Neuromatch Academy (2021): Methods and applications in computational neuroscience.
- Anti-Racist Pedagogy Learning Community, UC San Diego Teaching + Learning Commons (2020).
- Neurosurgery observation, UC San Diego Health (2018).